

6. Is the solution tailored for your use case?

Most of the AI solutions claim to be personalised for the end-users. However, the critical question is—before you deploy or start implementing those solutions—are these customised or tailored for you?

For many use cases, geography-specific rules apply. For example, if you are a European business, your obligations related to data handling and explainability, verifiability, etc. are different as compared to other countries. If your business is in China, facial recognition technology may be acceptable; however, for most of the other countries, it would be quite the opposite.

If you expect your solution to transcribe doctors' notes and make prognosis or decisions based on that, your solution must understand local vernacular, slangs, and other language nuances. For example, trash can (US) versus garbage bin (AU), or sidewalk (US) versus footpath (AU), mean the same things but are different words. The examples given here are quite rudimentary. However, for a sophisticated application, language differences would mean a lot.

In case of a hiring solution or applicant tracking system (ATS), wouldn't it be relevant if the solution was trained on localised or geography-specific data during the initial stage? The hiring pattern and criteria are significantly different for different countries, which means that the use of a standard solution is not a good fit for the purpose.

Geographical differences influence user interactions with any software system, and so does the user experience. It can be best handled during the solution design phase if the development team has local experts on board. These experts can provide valuable insights and knowledge to make the solution more effective and adaptable to the use case by end-users.

Not having a tailored solution for your use case is not a deal-breaker in most cases. However, it is safer to assume that the relevance is essential, so think about it, and evaluate it.

If the solution is not tailored to your use case, you may be dealing with

high latent risk. Evaluating the level of customisation can help in estimating potential risks.

7. Is the solution equipped to handle drift?

The problem of change in data over time and thereby affecting statically programmed (or assumed) relationships is a common occurrence for several other real-life scenarios of machine learning. The technical term, in the field of ML, for this scenario is 'concept drift.'

Here the term concept refers to the unknown or hidden relationship between inputs and outputs. The change in data can occur due to various real-life scenarios, which can result in degraded performance of your AI solution.

If the solution is designed right, it would have accounted for such drift in the first place and therefore would be able to adapt to the changing scenarios. If that isn't the case, either your solution would fail, or the performance will degrade. It may also mean lower accuracy, lower speed, high error margins, and other such issues.

However, remember that it is difficult to identify any scenario in which concept drift may occur, especially when you are training the model for the first time. So, instead of working on pre-deployment detection, your solution should safely assume that the drift will occur and accordingly have a provision to handle it.

If your solution is not capable of handling drifts, you will have to fix it eventually on an ongoing basis. It is a costly proposition as you keep spending money on constant fixes. The risk of tolerating poor performance for some time exists. This risk increases if the detection of drift is not apparent. If you, for some reason, could not detect the drift for a more extended period, it can be quite risky on several fronts. Thus, the

detection of drift is a critical factor too.

Check if your solution has accounted for concept drift and if yes, then to what extent? The answer to that check would tell you the level of risk.

A thorough evaluation is critical.

From an economic point of view, there is significant information asymmetry that has put AI solution creators in a more powerful position than its users.

It is being said that technology buyers are at a disadvantage when they know much less about the technology than sellers do. For the most part, decision-makers are generally not equipped to evaluate these systems.

The crux of the matter is, you cannot leave it to the AI developers, in-house or outside, to police themselves for mindful design and development.

It is natural for us to evolve and keep pushing the envelope. However, certain things cannot be undone. There is no *undo* option here. Therefore it is imperative that we either slow down or be overly cautious.

Since slowing down is a less likely acceptable option, being overly cautious is necessary. Being strict in the evaluation of risks and putting up strong governance are some of the critical steps you can take.

If your search engine shows irrelevant results or a movie streaming service suggests flop movies, it still may be a minor nuisance. However, if the system is controlling a car, or a power grid, or a financial trading system, it better be right, as even minor glitches are unacceptable.

For you to be confident in your AI system, you must make sure that you are working on the right solution, and the solution is designed right.

Tactically, 'doing things right' is essential, while strategically, 'doing the right things' is critical. But, with a powerful technology like AI at hand, both are essential. 

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